

TwinFusion AI

Backend Integration & API Guide

Project:	TwinFusion AI Freshers Portal
Track:	Google Student Ambassador Hackathon
Tech Stack:	FastAPI, SQLite, React, Google Gemini API
Document:	Step-by-step Authentication & Setup Walkthrough

1. Understanding the Architecture

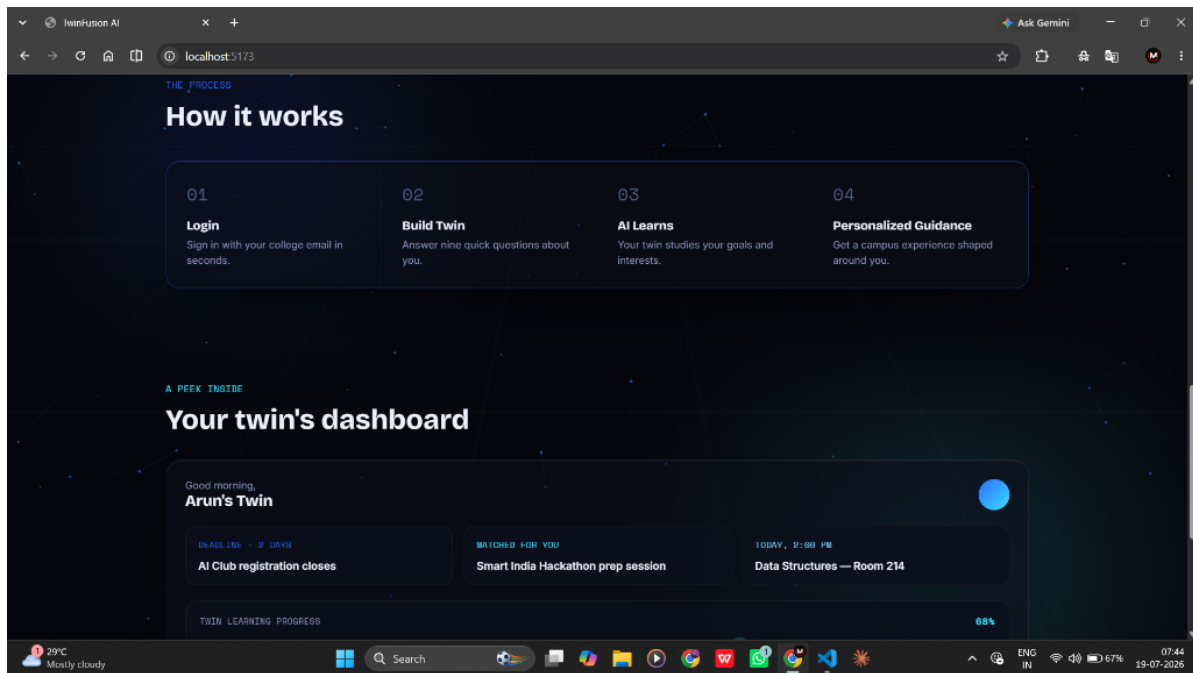
To understand how the application handles requests, let's use the simple **Restaurant Analogy**:

Component	The Restaurant Equivalence
Frontend (React)	The Dining Room: Where customers sit, browse the visually designed menu, and place orders. In demo mode, orders are mocked for speed.
Backend (FastAPI)	The Kitchen: The busy backroom where chefs receive slips, cook logic, secure keys, and process operations.
Database (SQLite)	The Pantry / Fridge: Where accounts, profiles, and scheduled calendars are stored. If the fridge is empty, the kitchen cannot serve the customer.
Google Gemini	The Secret Recipe Book: The cognitive advisor used by chefs to compose highly smart and customized recommendations.

***Why did 'Unauthorized' happen?** Since you booted the backend database for the first time, your database (the pantry) was completely empty. There were no user accounts inside it. When you typed credentials into the Swagger auth page, the kitchen searched the pantry, found nothing, and rejected the entry.*

2. Resolving Swagger Authentication

Here is the screenshot of the authorization modal where the error occurred:



Step-by-Step Fix to Authenticate Successfully:

Step 1: Write an Account to the Database first

Scroll down to the **auth** section in the Swagger documentation page. Expand the **POST /api/v1/auth/register** endpoint, click 'Try it out', and execute it with a request body like this:

```
{ "email": "student.tarun@college.edu", "username": "starun", "password": "Tarun@28",  
  "full_name": "S Tarun" }
```

Step 2: Log in with the Authorize lock button

Once the account is registered (returning 200 OK), click the green **Authorize** lock button on the top right. Enter your username (starun) and password (Tarun@28) and click Authorize. It will lock successfully!